Huawei Access Node BAS-ACC-07 Hardware/Environmental Fault Resolution Procedure

Node: BAS-ACC-07 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_9024

Description

The Huawei Access Node BAS-ACC-07, located at the Basavanagudi POP, is susceptible to Hardware/Environmental Faults due to the inherent design of the ASIC memory parity error detection mechanism, line card crash recovery procedures, and power supply redundancy configuration. The Access Node's Role in the network is to facilitate seamless data transmission between the core network and the user's premises. The failure domain of Hardware/Environmental Faults can be attributed to the following: (i) ASIC memory parity errors: The Access Node's ASIC (Application-Specific Integrated Circuit) is responsible for processing and forwarding data packets. In the event of a memory parity error, the ASIC may crash, leading to a node failure. (ii) Line card crashes: The line card is a critical component of the Access Node, responsible for transmitting and receiving data packets. In the event of a line card crash, the node will fail to forward data packets, resulting in a network outage. (iii) Power supply redundancy loss: The Access Node is equipped with redundant power supplies to ensure continuous operation in the event of a power supply failure. However, if both power supplies fail, the node will shut down, resulting in a network outage. The Alarm Code ALARM_9024 indicates a Major Hardware/Environmental Fault, which requires immediate attention to prevent network downtime.

Pre-requisite Checks

- pre_requisite: Verify the Basavanagudi POP site location and ensure remote hands are dispatched to the site.

- pre_requisite: Check the redundant link status to ensure it is operational and functioning correctly.

- pre_requisite: Verify the power supply redundancy configuration to ensure it is correctly configured and functioning as expected.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display fault-managed node fault-info, Expected Output: Node ID: BAS-ACC-07, Site Location: Basavanagudi POP, Vendor: Huawei, Role: Access, Failure Domain: Hardware/Environmental Fault, Alarm Code: ALARM_9024, Description: This command is used to display the node fault information, including the node ID, site location, vendor, role,
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failure domain, and alarm code.

Command: display fault-managed node fault-info detail, Expected
Output: Node ID: BAS-ACC-07, Site Location: Basavanagudi POP, Vendor:
Huawei, Role: Access, Failure Domain: Hardware/Environmental Fault,
Alarm Code: ALARM_9024, Fault Type: ASIC memory parity error,
Description: This command is used to display the node fault
information in detail, including the fault type and description.

Command: display system redundancy power-supply, Expected Output:
Redundant power supply status: Active/Active, Description: This
command is used to display the redundant power supply status,
including the active and standby power supplies.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: replace {board} {asic}, Description: This command is used to replace the faulty ASIC with a new one. Please ensure to replace the entire board if the ASIC is faulty.

Step 2: Step: Replace the faulty line card, Command: replace {board} {line-card}, Description: This command is used to replace the faulty line card with a new one. Please ensure to replace the entire board if the line card is faulty.

Step 3: Step: Verify the power supply redundancy configuration, Command: display system redundancy power-supply, Description: This command is used to verify the power supply redundancy configuration and ensure it is correctly configured and functioning as expected.

Post-Check Verification

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Command: display system redundancy power-supply, Expected Output:
Redundant power supply status: Active/Active, Description: This
command is used to verify the redundant power supply status, including
the active and standby power supplies.

Command: display fault-managed node fault-info, Expected Output: Node
ID: BAS-ACC-07, Site Location: Basavanagudi POP, Vendor: Huawei, Role:
Access, Failure Domain: Hardware/Environmental Fault, Alarm Code:
ALARM_9024, Description: This command is used to verify the node fault
information, including the node ID, site location, vendor, role,
failure domain, and alarm code.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: restore {config-file}, Description: This command is used to restore the original configuration from the backup file. Please ensure to back up the current configuration before making any changes.

- Step: Replace the faulty power supply, Command: replace {power-supply}, Description: This command is used to replace the faulty power supply with a new one. Please ensure to replace the entire power supply if it is faulty.